

EDUCATION

- **The University of Michigan** Ann Arbor, MI
Computer Science B.S.E. 2014 – 2018

EXPERIENCE

- **Michigan Medicine** Ann Arbor, MI
Software Engineer 2023 – Present
 - **Digital Pathology Tools:** Design, develop, and maintain the frontend, backend, and DevOps of a variety of tools, applications, and dashboards to improve digital pathology workflows using PHP, React, SQL, and Jenkins, increasing efficiency of labs, aiding in the digitization of pathology operations, and giving insights into process improvement opportunities. Currently leading the migration to a modern OpenShift stack using Laravel, TypeScript, and Postgres, improving scalability and separation of concerns by decoupling from our suite of lab applications
 - **Virtual Slide Box Rewrite:** Modernized a legacy slide viewer by analyzing the existing PHP and JavaScript codebase, then rebuilt it from the ground up in Go, React, and Postgres using iipsrv to serve high-magnification whole slide images. Deployed in OpenShift, improving scalability and maintainability
 - **Whole Slide Image Tilesource:** Contributed a custom tilesource to OpenSeadragon, an open-source web-based viewer for high-resolution zoomable images, specifically geared toward improving image viewing performance in digital pathology
 - **Sectra Integration & Support:** Partnered with key stakeholders and the Sectra team to deliver solutions including real-time image transfer and audit and an integrated information window in the desktop application, while providing ongoing support and troubleshooting to minimize downtime and ensure reliability
 - **Slide Deidentification:** Overhauled a decades-old slide deidentification script using modern technologies and languages including Python, Docker, Jenkins, and Git to improve speed, reliability, and maintainability
 - **Scanner Dashboard:** Developed a web application using PHP, React, and SQL to monitor and manage a fleet of high throughput slide scanners, including real-time status updates and error notifications, allowing for increased efficiency of lab operations and reduced scanner downtime
 - **Internal Tools & Automation Engineering:** Build and maintain internal tools for data analysis, process automation, and workflow optimization, replacing manual processes with scalable solutions using PHP, React, Python, SQL, and Jenkins
- **Coupa Software, Inc.** Ann Arbor, MI
Software Engineer 2023
 - **Refactoring & Quality Engineering:** Analyzed and updated the Kafka custom connector, a critical platform component for loading data to and from JDBC-compatible databases. Refactored code using design patterns and recommendations from Sonar, increased unit test coverage from 20% to 80%, and improved documentation all while ensuring functionality was maintained
- **Nutshell, Inc.** Ann Arbor, MI
Software Engineer 2022
 - **CRM Enhancements & External Integrations:** Developed CRM features including multifactor authentication using PHP, GraphQL, and SQL. Integrated customer data and APIs with an acquiring firm, enabling a seamless transition
- **Domino's Pizza, LLC** Ann Arbor, MI
Software & DevOps Engineer 2018 – 2022
 - **Kubernetes Platform Engineering:** Implemented Pivotal Kubernetes Service (PKS) within the environment, which involved cluster buildout, environment configuration, monitoring, Helm chart development, and infrastructure-as-code
 - **CI/CD & Automated Validation:** Built out JenkinsX within the PKS environment using Helm charts and running end-to-end validation scripts for each update to ensure the environment was stable and contained the latest changes, allowing for more rapid development
 - **Java Installation Automation:** Developed a method for seamless and automated Java installation for runtime environments, cutting time spent by SRE resources installing Java on VMs. The change empowered development teams to more easily upgrade the JDK version for their own services, increasing the security and reliability of their applications
 - **Digital Shoulder Surfing:** Working on a lean team as the sole backend developer, used Java, Spring framework, Maven, ActiveMQ and other tools to develop the backend service for a technology which enables in-store employees to see order information ahead of time, allowing them to prepare orders more quickly
 - **Service Health Tool:** Built a service health tool to validate versions, detect failures during releases, and automatically restart downed services, improving resiliency and reducing manual intervention

SKILLS

- **Programming:** Python, PHP, Go, Java, C++, C#, Bash, SQL, Postgres, R, Ansible, Puppet
- **Technologies:** Kubernetes, OpenShift, Docker, Helm, Jenkins, Nginx, .NET, Linux, Windows, Visual Studio, Git, GraphQL
- **Web Development:** React, HTML, CSS, JavaScript, TypeScript, Laravel, Axios, jQuery